

AUYUSH GANAPATHY

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| <https://github.com/Auyushg> | Portfolio: <https://auyushg.github.io>

EDUCATION

The University of Texas at Austin, Austin, TX

August 2025 - December 2028

Bachelor of Science, Computational Engineering - GPA: 3.83

Relevant Coursework: Dynamics, Advanced Calculus, Differential Equations with Linear Algebra, Engineering Computation

Work Eligibility: Permanent Resident

EXPERIENCE

CONNECTION HELIX CENTER FOR AI AND ROBOTICS, Merrimack, NH(Remote)

October 2025 - April 2026

Technical Intern

- Implemented digital twin robotic simulations in Isaac Sim / Isaac Lab on Ubuntu 22.04 to support autonomy and perception development
- Trained a reinforcement learning policy using Proximal Policy Optimization (PPO) on a simulated Unitree Go2 quadruped using RSL-RL, achieving walking gait coordination across 12 joints through a 3-phase curriculum learning approach
- Developed an autonomous navigation system combining a rule-based waypoint planner with a learned locomotion policy and LiDAR-based obstacle avoidance
- Tested model by simulating robot navigation of 29 sequential waypoints through a USD warehouse environment, including narrow corridors and sharp turns
- Integrated a heuristic signal using waypoint lateral and yaw correction with LiDAR scans for robot movement guidance
- Generated ~1400 synthetic RGB-D datasets to train YOLOv8 object detection models for a simulated Unitree Go2 quadruped
- Deployed perception models into ROS2 pipeline, subscribed to simulated RGB camera data, and applied 40-frame temporal filtering to reduce detection noise
- Computed detected object's position using depth data and camera intrinsics, transformed coordinates into global reference frame using TF
- Implemented a ROS2 bridge integrating Isaac Lab simulation data with the object detection module, publishing odometry, position, RGB 8-bit data, and depth camera topics
- Visualized and validated object detection outputs using RVIZ2

NASA JOHNSON SPACE CENTER, Houston, TX

June 2025 - August 2025

ER4 Robotics Intern

- Developed Python scripts using LiDAR and ROS2 to read scans, compute distance to surrounding objects, and detect objects in 20 cm proximity
- Collaborated with NASA engineers for LiDAR implementation on rover "MicroChariot", contributing to perception and autonomous path-planning development

PROJECTS

PINN-BASED 6-DOF ROCKET TRAJECTORY ANALYSIS & SIMULATION

May 2026 – Present

- Built a 6-DOF rocket flight simulator involving quaternion kinematics and RK4 integration, modeling thrust, mass depletion, and attitude control for multiple vehicles
- Generated a 50-run Monte Carlo dataset ($\pm 10\%$ dispersions on initial state and control gain multipliers) across 5 vehicle scenarios, producing ~10M+ training samples
- Trained a PyTorch-based physics-informed neural network to predict state derivatives from state, control, and vehicle-parameter inputs, implementing hyperbolic tangent nonlinearity
- Regularized training with a physics residual loss enforcing 6-DOF equations at randomly sampled collocation points (state perturbed $\pm 10\text{-}25\% \sigma$), with direct supervision against simulated derivative data
- Achieved up to 79.76% accuracy on quaternion predictions
- Diagnosing training instabilities from cross-vehicle gradient interference, testing hyperparameters, and applying cosine similarity diagnostics to guide loss-term rebalancing

LEADERSHIP & COMMUNITY INVOLVEMENT

LONGHORN ROCKETRY ASSOCIATION, Austin, TX

September 2025 – Present

Competition Team *Simulation Engineer*

- Co-developed a 6-DOF flight simulation in Python using RocketPy to model a 3.66 m rocket's trajectory, incorporating mass properties and aerodynamic drag
- Designed and simulated a PD-based airbrake control system using numerical apogee estimation and correction to regulate flight altitude to target constraints (~10000 ft AGL)
- Modeled airbrakes and involved nonlinear aerodynamic behavior by implementing Mach speed and deployment-dependent drag coefficient surfaces for speeds up to Mach 0.93
- Implemented wyoming sounding atmospheric model, involved wind speeds and atmospheric density in trajectory prediction

ADDITIONAL SKILLS

Technical Skills: Java, Python, C++, PID, Matplotlib, Lua, Matlab, ROS2, YOLOv8 Object Detection, IsaacLab, Git, Ubuntu 22.04, PyTorch, Reinforcement Learning AI, Numpy, Physics-Informed Neural Networks, 6-DOF Flight Dynamics